

Report R10 — Permutation circuits: the same 256 rules with $V = X$

A control for the sector/attractor machinery, open boundaries

QCA Sector Analysis project (Tier 1)

2026-07-30 (xgate_version 1x.1, obc0)

Abstract

The Tier-1 machinery separates two decompositions of the computational basis: the weakly connected components (sectors, exact for both the monitored and the unmonitored dynamics) and the terminal strongly connected components (monitored attractors). In the Hadamard family those two disagree strongly, and R9 is largely about the disagreement. This report runs the identical analysis on the same 256 symbol tables with the active gate changed from the Hadamard to X . Every local symbol is then a function on bits, the transition graph has out-degree one, and three things follow at once: the two decompositions become *identical* in number, sectors and basins coincide, and the 16 rules built from I and V alone are reversible cellular automata whose sectors are exactly the permutation cycles. We verify all of this over 256 rules, confirm that the partition sum rule and the exclusion curve carry over unchanged, and record the one place where the two families genuinely differ in physics rather than in bookkeeping: the permutation circuits fragment into exponentially many, exponentially large sectors whose *attractors are almost always short*. Only 24 of 256 rules have an exponentially long cycle, and for the 80 V -free rules the number is zero. Swept to $N = 24$ for all 256 rules. obc0 only; pbc is held.

Contents

1	What changes when $V = X$	1
1.1	What “reversible” means here, and what it does not	2
2	Coverage	3
3	The headline: large sectors, short attractors	3
4	The reversible rules	3
5	What carries over, and what does not	5
6	Validation	6
7	Open items	6
8	Reproduce	7

1 What changes when $V = X$

The symbol table is untouched. Each entry still says what happens at a site for one neighbour pattern; only the active gate is different:

symbol	Hadamard family (R1–R9)	this report
I	identity	identity
V	Hadamard	X , i.e. $x_i \mapsto 1 - x_i$
D	reset to $ 0\rangle$	reset to $ 0\rangle$
E	reset to $ 1\rangle$	reset to $ 1\rangle$

All four are now *functions* on the computational basis, so the circuit is one too, and $\text{succ}(x)$ is a single state. Three consequences, and they are the reason this family is a useful control.

Proposition 1 (functional graph). *The one-cycle transition graph has out-degree exactly 1.*

Theorem 1 (F1: one cycle per component). *In a functional graph every weakly connected component contains exactly one cycle. Hence for every X -gate rule, every N and every boundary condition,*

$$n_{\text{rec}} = n_{\text{wcc}}, \quad (1)$$

the basin of each cycle is its entire component, and no state lies in two basins.

That identity is exactly what fails in the Hadamard family, and fails badly: R9’s rule 22 at $N = 12$ has three monitored attractors, each a single state, sitting inside two sectors that cover all 4096 states. There A4 could only be asserted in one direction. Here the two tiers coincide by construction, so any disagreement in the code would show up immediately — which makes this family a test of the machinery as much as a physics question.

1.1 What “reversible” means here, and what it does not

The word is overloaded in this area, so it is fixed explicitly.

Definition 1. A rule is *reversible at* $(N, \mathbf{bc})$ if its one-cycle map $\{0, 1\}^N \rightarrow \{0, 1\}^N$ is a bijection.

Proposition 2. *A rule is reversible, for every $N \geq 3$ and both boundary conventions, iff its symbol table uses only I and V. There are $2^4 = 16$ such rules: 51, 54, 57, 60, 99, 102, 105, 108, 147, 150, 153, 156, 195, 198, 201, 204.*

Proof sketch. One elementary step reads the two neighbour bits of site i and then applies I or X to site i . The controls are bits $\neq i$ and are untouched by that step, so the step is a controlled- X and hence a bijection; a composition of bijections is a bijection, whatever the layer order. Conversely D and E send both values of the target bit to the same value, so two states differing only in site i but sharing its neighbour pattern collide. $\square$

Note the proof does not use the sublattice structure, so it survives the case where the even sites fail to be an independent set (**pbc** at odd N). Checked exhaustively anyway at $N = 7, 8, 9$ for both conventions: exactly those 16 are bijections and none of the other 240 is.

Not the six reversible ECA. A better-known count lives nearby and the two must not be conflated. For *elementary* cellular automata — one layer, simultaneous update, each cell a function of its three-cell neighbourhood — only **six** of the 256 rules are reversible: 15, 51, 85, 170, 204, 240, the shifts and complements.

Two things differ, not one. The construction differs: the brick-wall QCA updates one sublattice at a time and applies a gate to the target conditioned on its neighbours, which makes invertibility a purely *local* property of the gate. And the quantifier differs: the ECA six are those bijective for *every* N , an intersection over lattice sizes, and at a fixed N the bijective set is strictly larger — on a ring of $N = 7$ it also contains 45, 75, 89 and 150. Our 16 need no such intersection: the proposition above is proved locally and holds at every N separately, which is why the two counts are not comparable term by term. Only 51 and 204 lie in both sets.

So the 16 rules that were “unitary” in the Hadamard family are here *reversible* in the sense just defined: every state is recurrent, each component is a single cycle, and the sector decomposition *is* the orbit decomposition of a permutation of $\{0, 1\}^N$.

Implementation. The site order and the neighbour conventions are not reimplemented. The compiled step list comes straight from `core/cycle.py::_compile`, so the brick-wall order (even sites ascending, then odd, each reading the current state) and the boundary handling are byte-for-byte those of the validated engine; only the local action is swapped. Nothing in the Hadamard engine is touched, and the two live in separate stores. This matters because the brick-wall order is *not* two simultaneous sublattice updates once the even sites stop being an independent set.

Cost. The whole analysis is one $O(2^N)$ pass over a functional graph — a three-colour walk that finds the cycles and, on the way back, each node’s distance to one. No amplitudes, no Tarjan stack, no union-find needed (though we run union-find anyway as a cross-check, §6). All 256 rules at $N = 24$ — sixteen million states each — take 43 min in total, against hours per *rule* for the Hadamard sweep at $N = 16$. The map itself is built with numpy, lifting the loop over x out of Python while keeping the site order from `_compile`, which is what makes $N = 24$ affordable; the vectorised and scalar builders are checked against each other in the tests.

2 Coverage

N	rules covered	2^N
6	256	64
7	256	128
8	256	256
9	256	512
10	256	1 024
11	256	2 048
12	256	4 096
13	256	8 192
14	256	16 384
15	256	32 768
16	256	65 536
17	256	131 072
18	256	262 144
19	256	524 288
20	256	1 048 576
21	256	2 097 152
22	256	4 194 304
23	256	8 388 608
24	256	16 777 216

Uniform over all 256 rules. `pbk` is deliberately not run: the ring has its own caveats and is held for a separate report.

3 The headline: large sectors, short attractors

family	rules	exp. sectors	exp. cycles	median cyclic frac.	median depth
reversible	16	6	6	1	0
V+reset	160	155	18	1.79×10^{-7}	13
V-free	80	78	0	1.46×10^{-5}	12

Read the two middle columns against each other. **239 of 256 rules have exponentially large sectors, and only 24 have exponentially long cycles.** For the 80 V-free rules the count is *zero*: not one of them has a cycle whose length grows exponentially, while 78 of them have exponentially large sectors. The median cyclic fraction — the share of the basis that is periodic at all — is 1.8×10^{-7} for the V+reset family and 1.5×10^{-5} for the V-free family at $N = 24$, with a median transient depth of 13 and 12 steps.

So the permutation circuits are the textbook classical picture: a state falls through a long transient into a short attractor, and essentially all of the Hilbert space is transient. The sector structure is large and the recurrent structure is tiny, and the gap between them is the whole content of the family.

X1 X-gate circuits: the two maps share their x-axis exactly, because a functional graph has one cycle per weak component (obc0).

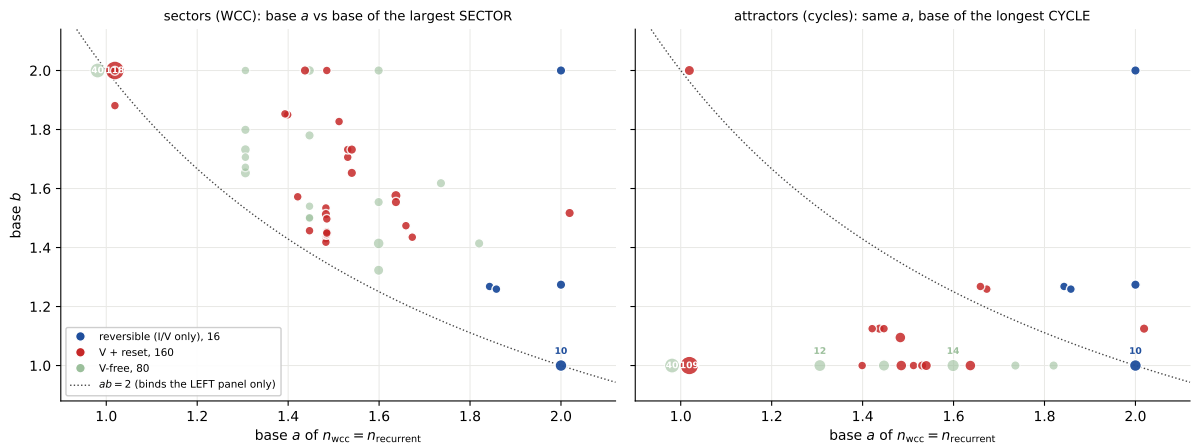


Figure 1: **X1, the two maps.** Left: base of the sector count against base of the largest *sector*. Right: the same x -axis — identical by (1), not merely similar — against the base of the longest *cycle*. The dotted curve is $ab = 2$; it binds the left panel, because sectors partition, and is only a reference on the right, because cycles do not. The collapse of the right panel onto $b = 1$ is the headline: marker area grows with the number of rules in a cell and the two largest cells there hold 110 and 40 rules.

4 The reversible rules

For the 16 rules built from I and V the two decompositions are not just equal in number: the sectors *are* the cycles, elementwise. So a single table describes both maps.

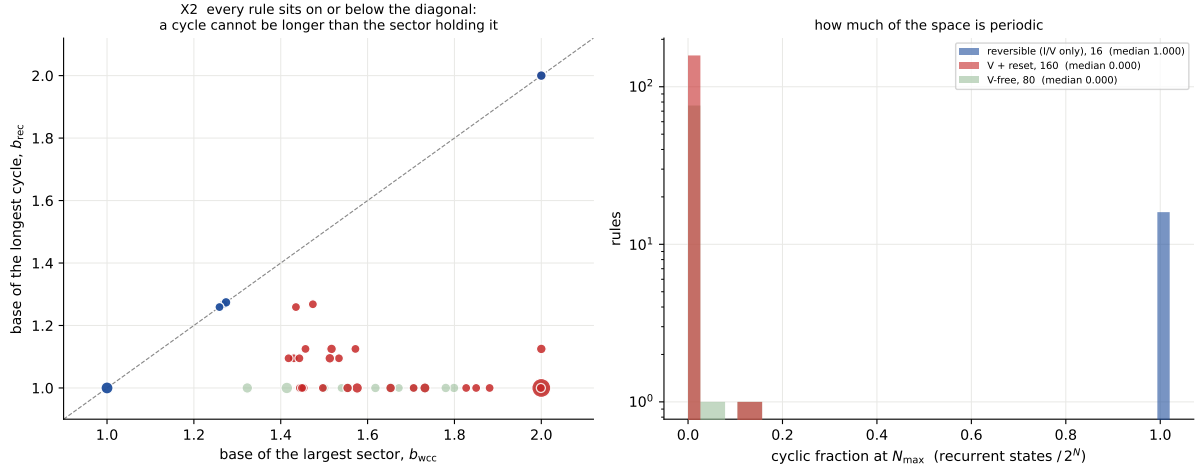


Figure 2: **X2, the drop, per rule.** Left: longest cycle against largest sector. Every rule lies on or below the diagonal — a cycle cannot be longer than the sector containing it — and the reversible rules are exactly the ones *on* it, since for them the sector *is* the cycle. Right: the distribution of the cyclic fraction, log-scaled, by family.

rule	tuple	a	b	ab	n_{wcc}	D_{max}
51	VVVV	2.00000	1.00000	2.0000	8 388 608	2
54	IVVV	2.00000	1.00000	2.0000	364 724	67
57	VIVV	2.00000	2.00000	4.0000	1 798	6 807 384
60	IIVV	2.00000	1.00000	2.0000	526 356	32
99	VVIV	2.00000	2.00000	4.0000	1 798	6 807 384
102	IVIV	2.00000	1.00000	2.0000	526 356	32
105	VIIV	2.00000	1.00000	2.0000	671 092	25
108	IIIV	1.84296	1.26792	2.3367	2 373 460	391
147	VVVI	2.00000	1.00000	2.0000	364 724	46
150	IVVI	2.00000	1.00000	2.0000	671 092	25
153	VIVI	2.00000	1.00000	2.0000	524 288	32
156	IIVI	2.00000	1.27441	2.5488	1 170 866	420
195	VVII	2.00000	1.00000	2.0000	524 288	32
198	IVII	2.00000	1.27441	2.5488	1 170 866	420
201	VIII	1.85800	1.25932	2.3398	1 812 802	561
204	IIII	2.00000	1.00000	2.0000	16 777 216	1

Three anchors are exact and worth stating, because they are the analogues of R9's:

- **Rule 204 (IIII)** is the identity: 2^N fixed points, $a = 2$, $b = 1$, $ab = 2$ — on the curve, as in the Hadamard case, but for a different reason (there every sector was a frozen singleton; here every *cycle* is).
- **Rule 51 (VVVV)** is the global flip $x \mapsto \bar{x}$: 2^{N-1} two-cycles, so $a = 2$, $b = 1$ and $ab = 2$. Note the contrast with the Hadamard family, where 51 gives *one* sector of size 2^N — the same symbol table, opposite extremes of the sector axis, purely from the gate.
- **Rule 150 (IVVI)** with $V = X$ is the linear map $x_i \mapsto x_i \oplus x_{i-1} \oplus x_{i+1}$ applied brick-wall. Its longest cycle is short and grows *linearly*: $D_{max} = 14, 15, 16, 17, 18$ at $N = 13 \dots 17$, i.e. $N + 1$ over the range computed, with $n_{wcc} \approx 2^N / (N + 1)$ so $a = 2$, $b = 1$, $ab = 2$. Against the Hadamard rule 150, whose sectors are the binomial domain-wall shells of R8, this is a completely different object built from the same table.

The family spans the whole plane. At one end 204 and 51 have 2^N and 2^{N-1} tiny orbits; at the other, 57 and 99 (VIVV, VVIV) have only 110 sectors at $N = 17$ with the largest holding 36 456 states, giving $a = 1.405$, $b = 1.961$ and $ab = 2.755$. Four rules — 60, 102, 153, 195 — and also 108 and 201 sit at $b = 2$ exactly with $a \approx 1.79$ – 1.85 , so $ab \approx 3.6$ – 3.7 , far above the exclusion curve: many orbits *and* a largest orbit that is a fixed fraction of the space.

5 What carries over, and what does not

Carries over. The sum rule and the exclusion curve are properties of a *partition*, and sectors partition here exactly as they did there. The fit-free finite- N form

$$n_{\text{wcc}}(N) \cdot D_{\text{max}}(N) \geq 2^N$$

holds for all 256 rules at every N , and is tight in the same place — the smallest ratio anywhere is exactly 1.0000, at rule 0, $N = 6$.

Does not carry over. Everything that depended on the Hadamard being a *superposing* gate. There is no interference, so no destructive cancellation in succ; no dark states; no distinction between the monitored and the unmonitored channel at the level of the support, since a single Kraus branch survives on each basis state. The Tier-1d monitored/unmonitored certificate is vacuous here. Conversely the transient structure, which was a thin and somewhat awkward part of the Hadamard story, becomes the dominant feature.

The one identity that is a genuine control. $n_{\text{rec}} = n_{\text{wcc}}$ held for **every unit computed**, with zero failures, and the union-find sector decomposition — which knows nothing about functional graphs — reproduced the basin decomposition exactly wherever both were run. Two independent code paths, one theorem, no discrepancies.

X3 transient structure of the permutation circuits (obc0); the reversible rules sit at cyclic fraction 1 and depth 0 by construction.

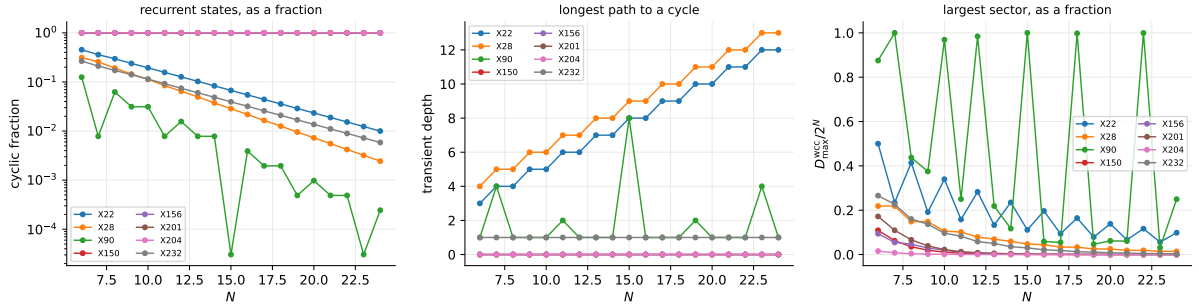


Figure 3: **X3, transient structure.** Cyclic fraction (log scale), transient depth, and largest sector as a fraction of the space, against N . The reversible rules sit at cyclic fraction 1 and depth 0 by construction; the irreversible ones fall away exponentially.

6 Validation

check	applicable	passing	note
sector sizes partition, $\sum_k D_k = 2^N$	4864	4864	asserted in the analysis
$n_{\text{recurrent}} = n_{\text{wcc}}$ (F1)	4864	4864	0 failures
finite- N $n_{\text{wcc}} D_{\text{max}} \geq 2^N$	256	256	tightest 1.0000 at $N = 6$
sector map $ab \geq 2$	249	247	2 below
cycle map $ab \geq 2$ (<i>not</i> required)	241	24	217 below, as expected

The last row is the point of the family rather than a failure: the cycle map has no exclusion curve, because cycles do not partition the basis, and 217 of 241 rules do sit below $ab = 2$ there. In the sector map, where the curve does bind, only *two* rules fall below — and neither is a violation. Both are the degenerate case R9 §5 documents, where the sector count is sub-exponential so the base product silently drops the polynomial factor: X_{25} (VIVD) has an essentially constant sector count (2, 3, 2, 2, 3, 2 over $N = 19 \dots 24$) with $b = 1.99887$, and X_{182} (IVVE) a linear one (11, 11, 12, 12, 13, 13) with $b = 1.881$. With $a = 1$ the inequality degenerates to $b \geq 2$, approached from below at finite N , which is what both are doing. Carrying the sweep from $N = 20$ to $N = 24$ took this count from 14 to 2, so the residue is a finite-size artefact and not a family of offenders.

7 Open items

1. **pbc.** Held, as for R9.
2. **Rule 150's cycle length.** $D_{\max} = N + 1$ over $N = 13 \dots 17$ is five points; the map is GF(2)-linear, so the orbit structure is governed by the order of a matrix over GF(2) and should be derivable rather than fitted. That derivation is not attempted here.
3. **The 24 rules with exponentially long cycles.** Six are reversible (where a long cycle is unsurprising); the other eighteen are V+reset rules that nonetheless sustain an exponentially long attractor despite a median cyclic fraction of 10^{-7} . That list is short and deserves the same treatment R9 gave its eight open-system-fragmented rules.
4. **Comparison with the Hadamard sector map** rule by rule. The two families share a symbol table but not a single quantitative law; whether any rule-level correlation survives the gate change is untested.

8 Reproduce

```
python -m qca_fragmentation.permutation.sweep --bc obc0 --n-min 6 --n-max 20
python -m qca_fragmentation.permutation.analysis --bc obc0
python -m qca_fragmentation.permutation.tables --bc obc0
python -m qca_fragmentation.permutation.figures --bc obc0 --rebuild
python -m pytest qca_fragmentation/tests/test_xgate.py -q
```

Code: permutation/xca.py (the map, ported from core/cycle.py), permutation/functional.py (the one-pass decomposition and the union-find cross-check), permutation/analysis.py, permutation/tables.py, permutation/figures.py. Data: results_xgate/{rule}-{bc}.jsonl and analytics/xgate_map-obc0.json.